

Australian statement of hazardous nature : Classified as hazardous according to criteria of Safe Work Australia

## Section 1 - Identification

**Product Name** Pentadecafluorooctanoic acid

**CAS No** 335-67-1

**Product Code** 173960000; 173960050; 173960250; 173961000

**Address** ThermoFisher Scientific Australia Pty Ltd  
5 Caribbean Drive, Scoresby  
VICTORIA 3179, Australia

**Emergency Tel.** **CHEMTREC®**  
**03 9757 4559 or +613 9757 4559**

**Telephone / Fax Numbers** Tel: 1300 735 292

Fax: 1800 067 639

**E-mail address** ANZinfo@thermofisher.com

**Recommended Use** Laboratory chemicals.

**Uses advised against** This product does not contain any substance(s) on the Illicit Drug Precursors/Reagents list.  
This product does not contain any substance(s) subject to Prohibition, Authorization or Restriction. This product does not contain any substance(s) listed on the voluntary National Code of Practice for Chemicals of Security Concern.

## Section 2 - Hazard(s) Identification

### Classification under Safe Work Australia

Classified as hazardous according to criteria of Safe Work Australia

#### Physical hazards

No hazards identified

#### Health hazards

Acute Oral Toxicity

Category 4

Acute Inhalation Toxicity - Dusts and Mists

Category 4

Serious Eye Damage/Eye Irritation

Category 1

Carcinogenicity

Category 2

Reproductive Toxicity

Category 1B

/; Effects on or via lactation

Specific target organ toxicity - (repeated exposure)

Category 1

#### Environmental hazards

No hazards identified

#### Label Elements



Exclamation Mark



Health Hazard



Corrosion

**Signal Word****Danger****Hazard Statements**

H318 - Causes serious eye damage

H351 - Suspected of causing cancer

H360 - May damage fertility or the unborn child

H362 - May cause harm to breast-fed children

H372 - Causes damage to organs through prolonged or repeated exposure

H302 + H332 - Harmful if swallowed or if inhaled

**Precautionary Statements**

P264 - Wash face, hands and any exposed skin thoroughly after handling

P270 - Do not eat, drink or smoke when using this product

P201 - Obtain special instructions before use

P202 - Do not handle until all safety precautions have been read and understood

P260 - Do not breathe dust/fume/gas/mist/vapors/spray

P263 - Avoid contact during pregnancy and while nursing

P271 - Use only outdoors or in a well-ventilated area

P280 - Wear protective gloves/protective clothing/eye protection/face protection

P304 + P340 - IF INHALED: Remove person to fresh air and keep comfortable for breathing

P305 + P351 + P338 - IF IN EYES: Rinse cautiously with water for several minutes. Remove contact lenses, if present and easy to do. Continue rinsing

P310 - Immediately call a POISON CENTER or doctor

P330 - Rinse mouth

P403 - Store in a well-ventilated place

P501 - Dispose of contents/ container to an approved waste disposal plant

**Other information**

No information available

## Section 3 - Composition and Information on Ingredients

| Component                       | CAS No   | Weight % |
|---------------------------------|----------|----------|
| Octanoic acid, pentadecafluoro- | 335-67-1 | >95      |

## Section 4 - First Aid Measures

**Inhalation**

Remove to fresh air. If not breathing, give artificial respiration. Get medical attention if symptoms occur.

**Ingestion**

Clean mouth with water and drink afterwards plenty of water. Get medical attention if symptoms occur.

**Skin Contact**

Wash off immediately with plenty of water for at least 15 minutes. If skin irritation persists, call a physician.

**Eye Contact**

Rinse immediately with plenty of water, also under the eyelids, for at least 15 minutes. Get

|                                            |                                                                                                                                                                                                                                                                                                                   |
|--------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                            | medical attention.                                                                                                                                                                                                                                                                                                |
| <b>General Advice</b>                      | If symptoms persist, call a physician.                                                                                                                                                                                                                                                                            |
| <b>Self-Protection of the First Aider</b>  | Ensure that medical personnel are aware of the material(s) involved, take precautions to protect themselves and prevent spread of contamination.                                                                                                                                                                  |
| <b>First Aid Facilities</b>                | Eyewash, safety shower and washroom.                                                                                                                                                                                                                                                                              |
| <b>Most important symptoms and effects</b> | None reasonably foreseeable. Causes severe eye damage. Product is a corrosive material. Use of gastric lavage or emesis is contraindicated. Possible perforation of stomach or esophagus should be investigated: Ingestion causes severe swelling, severe damage to the delicate tissue and danger of perforation |
| <b>Notes to Physician</b>                  | Treat symptomatically.                                                                                                                                                                                                                                                                                            |

## Section 5 - Fire Fighting Measures

### Suitable Extinguishing Media

Water spray. Carbon dioxide (CO<sub>2</sub>). Dry chemical. Chemical foam.

### Extinguishing media which must not be used for safety reasons

No information available.

### Hazardous Decomposition Products

Carbon monoxide (CO), Carbon dioxide (CO<sub>2</sub>), Gaseous hydrogen fluoride (HF), Thermal decomposition can lead to release of irritating gases and vapors.

### Decomposition Temperature

> 300°C

### Specific Hazards Arising from the Chemical

Thermal decomposition can lead to release of irritating gases and vapors. In the event of fire and/or explosion do not breathe fumes.

### Special protective equipment and precautions for fire fighters

As in any fire, wear self-contained breathing apparatus pressure-demand, MSHA/NIOSH (approved or equivalent) and full protective gear.

## Section 6 - Accidental Release Measures

### Emergency procedures

Use personal protective equipment as required. Ensure adequate ventilation. Avoid dust formation.

### Environmental Precautions

Should not be released into the environment.

### Methods for Containment and Clean Up

#### Clean-up methods - small spillage

Sweep up and shovel into suitable containers for disposal. Keep in suitable, closed containers for disposal.

#### Clean-up methods - large spillage

Typically only supplied in small quantities as packaged goods.

If extremely toxic or used in larger quantities ensure a spillage action plan is in place. Evacuate area. Control the source and/or contain the spill if safe and able to do so. Use temporary diking, sand bags, dry sand, earth or proprietary booms/absorbent pads if available. Obtain advice on containment, neutralisation and clean-up from local emergency responders.

### Reference to Other Sections

Refer to protective measures listed in Sections 8 and 13.

## Section 7 - Handling and Storage

### Precautions for Safe Handling

Do not get in eyes, on skin, or on clothing. Wear personal protective equipment/face protection. Ensure adequate ventilation. Avoid ingestion and inhalation. Avoid dust formation.

### Conditions for Safe Storage, Including any Incompatibilities

Corrosives area. Keep containers tightly closed in a dry, cool and well-ventilated place. Keep in properly labeled containers.

AS/NZS 2243.10:2004, Safety in laboratories - Storage of chemicals

## Section 8 - Exposure Controls and Personal Protection

### Exposure limits

DE - MAK and BAT values of Hazardous Chemical Compounds in the Work Area. Published by German Research Foundation on July 1, 2011

| Component                       | Australia | New Zealand WEL | ACGIH TLV | The United Kingdom | Germany                                                                              |
|---------------------------------|-----------|-----------------|-----------|--------------------|--------------------------------------------------------------------------------------|
| Octanoic acid, pentadecafluoro- |           |                 |           |                    | TWA: 0.005 mg/m <sup>3</sup> (8 Stunden). MAK Höhepunkt: 0.04 mg/m <sup>3</sup> Haut |

### Biological limit values

This product, as supplied, does not contain any hazardous materials with biological limits established by the region specific regulatory bodies

| Component                       | Australia | New Zealand | European Union | United Kingdom | Germany                                                |
|---------------------------------|-----------|-------------|----------------|----------------|--------------------------------------------------------|
| Octanoic acid, pentadecafluoro- |           |             |                |                | Perfluorooctanoic acid: 5 mg/L serum (no restriction ) |

### Exposure Controls

#### Engineering Measures

Ensure adequate ventilation, especially in confined areas. Ensure that eyewash stations and safety showers are close to the workstation location.

Wherever possible, engineering control measures such as the isolation or enclosure of the process, the introduction of process or equipment changes to minimise release or contact, and the use of properly designed ventilation systems, should be adopted to control hazardous materials at source

### Personal protective equipment

#### Eye Protection

Goggles (Australian/New Zealand Standard AS/NZS 1337 - Eye protectors for Industrial applications)

#### Hand Protection

Protective gloves

| Glove material | Breakthrough time | Glove thickness | AUS/NZ Standard | Glove comments        |
|----------------|-------------------|-----------------|-----------------|-----------------------|
| Nitrile rubber | See manufacturers | -               | AS/NZS 2161     | (minimum requirement) |
| Neoprene       | recommendations   |                 |                 |                       |
| Natural rubber |                   |                 |                 |                       |
| PVC            |                   |                 |                 |                       |

Inspect gloves before use.

Please observe the instructions regarding permeability and breakthrough time which are provided by the supplier of the gloves. (Refer to manufacturer/supplier for information)

Ensure gloves are suitable for the task: Chemical compatibility, Dexterity, Operational conditions, User susceptibility, e.g. sensitisation effects, also take into consideration the specific local conditions under which the product is used, such as the danger of cuts, abrasion.

Remove gloves with care avoiding skin contamination.

|                                        |                                                                                                                                                                                                                                                                                                                                                                                                                    |
|----------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Skin and body protection</b>        | Long sleeved clothing                                                                                                                                                                                                                                                                                                                                                                                              |
| <b>Respiratory Protection</b>          | Use an AS/NZS 1716 approved respirator if exposure limits are exceeded or if irritation or other symptoms are experienced. To protect the wearer, respiratory protective equipment must be the correct fit and be used and maintained in line with AS/NZS 1715 on the use and maintenance of respiratory protective devices                                                                                        |
| <b>Recommended Filter type:</b>        | Particulates filter conforming to EN 143 (or AUS/NZ equivalent)                                                                                                                                                                                                                                                                                                                                                    |
| <b>Recommended half mask:-</b>         | Particle filtering: EN149:2001 (or AUS/NZ equivalent)<br>When RPE is used a face piece Fit Test should be conducted                                                                                                                                                                                                                                                                                                |
| <b>Hygiene Measures</b>                | Keep away from food, drink and animal feeding stuffs. When using do not eat, drink or smoke. Contaminated work clothing should not be allowed out of the workplace. Provide regular cleaning of equipment, work area and clothing. Avoid contact with skin, eyes or clothing. Remove and wash contaminated clothing and gloves, including the inside, before re-use. Wear suitable gloves and eye/face protection. |
| <b>Environmental exposure controls</b> | No information available.                                                                                                                                                                                                                                                                                                                                                                                          |

## Section 9 - Physical and Chemical Properties

### Information on basic physical and chemical properties

|                                                |                                 |                                          |
|------------------------------------------------|---------------------------------|------------------------------------------|
| <b>Appearance</b>                              | Off-white                       |                                          |
| <b>Physical State</b>                          | Solid                           |                                          |
| <b>Odor</b>                                    | pungent                         |                                          |
| <b>Odor Threshold</b>                          | No data available               |                                          |
| <b>pH</b>                                      | 2.6                             | 1g/l aq.sol., 20°C                       |
| <b>Melting Point/Range</b>                     | 53 - 60 °C / 127.4 - 140 °F     |                                          |
| <b>Softening Point</b>                         | No data available               |                                          |
| <b>Boiling Point/Range</b>                     | 189 - 192 °C / 372.2 - 377.6 °F | @ 760 mmHg                               |
| <b>Flash Point</b>                             | No information available        | <b>Method -</b> No information available |
| <b>Evaporation Rate</b>                        | Not applicable                  | Solid                                    |
| <b>Flammability (solid,gas)</b>                | No information available        |                                          |
| <b>Explosion Limits</b>                        | No data available               |                                          |
| <b>Vapor Pressure</b>                          | No data available               |                                          |
| <b>Vapor Density</b>                           | Not applicable                  | Solid                                    |
| <b>Specific Gravity / Density</b>              | No data available               |                                          |
| <b>Bulk Density</b>                            | No data available               |                                          |
| <b>Water Solubility</b>                        | 3.4g/l                          |                                          |
| <b>Solubility in other solvents</b>            | No information available        |                                          |
| <b>Partition Coefficient (n-octanol/water)</b> |                                 |                                          |
| <b>Autoignition Temperature</b>                | No data available               |                                          |
| <b>Decomposition Temperature</b>               | > 300°C                         |                                          |
| <b>Viscosity</b>                               | Not applicable                  | Solid                                    |
| <b>Explosive Properties</b>                    | No information available        |                                          |
| <b>Oxidizing Properties</b>                    | No information available        |                                          |
| <b>Other information</b>                       |                                 |                                          |
| <b>Molecular Formula</b>                       | C8 H F15 O2                     |                                          |
| <b>Molecular Weight</b>                        | 414.07                          |                                          |

## Section 10 - Stability and Reactivity

|                   |                                            |
|-------------------|--------------------------------------------|
| <b>Reactivity</b> | None known, based on information available |
| <b>Stability</b>  | Stable under normal conditions.            |

|                                         |                                                                                                                                                                    |
|-----------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Conditions to Avoid</b>              | Incompatible products, Exposure to air or moisture over prolonged periods.                                                                                         |
| <b>Incompatible Materials</b>           | Bases, Strong acids, Reducing Agent.                                                                                                                               |
| <b>Hazardous Decomposition Products</b> | Carbon monoxide (CO). Carbon dioxide (CO <sub>2</sub> ). Gaseous hydrogen fluoride (HF). Thermal decomposition can lead to release of irritating gases and vapors. |
| <b>Hazardous Polymerization</b>         | No information available.                                                                                                                                          |

## Section 11 - Toxicological Information

### Information on Toxicological Effects

#### Product Information

**(a) acute toxicity;**

|                   |                   |
|-------------------|-------------------|
| <b>Oral</b>       | Category 4        |
| <b>Dermal</b>     | No data available |
| <b>Inhalation</b> | Category 4        |

| Component                       | LD50 Oral                     | LD50 Dermal | LC50 Inhalation |
|---------------------------------|-------------------------------|-------------|-----------------|
| Octanoic acid, pentadecafluoro- | LD50 200 - 2000 mg/kg ( Rat ) |             |                 |

**(b) skin corrosion/irritation;** No data available

**(c) serious eye damage/irritation;** Category 1

**(d) respiratory or skin sensitization;**

|                    |                   |
|--------------------|-------------------|
| <b>Respiratory</b> | No data available |
| <b>Skin</b>        | No data available |

**(e) germ cell mutagenicity;** No data available

**(f) carcinogenicity;** Category 2

The table below indicates whether each agency has listed any ingredient as a carcinogen

| Component                       | Australia | New Zealand | New South Wales | Western Australia | IARC     | EU | UK | Germany |
|---------------------------------|-----------|-------------|-----------------|-------------------|----------|----|----|---------|
| Octanoic acid, pentadecafluoro- |           |             |                 |                   | Group 2B |    |    |         |

**(g) reproductive toxicity;** Category 1B  
**Reproductive Effects** Experiments have shown reproductive toxicity effects on laboratory animals

**(h) STOT-single exposure;** No data available

**(i) STOT-repeated exposure;** Category 1

**Target Organs** Liver.

**(j) aspiration hazard;** Not applicable  
Solid

**Symptoms / effects, both acute and delayed** Product is a corrosive material. Use of gastric lavage or emesis is contraindicated. Possible perforation of stomach or esophagus should be investigated: Ingestion causes severe swelling, severe damage to the delicate tissue and danger of perforation

## Section 12 - Ecological Information

|                                        |                                                                                                                                                             |
|----------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Ecotoxicity effects</b>             | Do not flush into surface water or sanitary sewer system. Do not allow material to contaminate ground water system.                                         |
| <b>Persistence and Degradability</b>   |                                                                                                                                                             |
| <b>Persistence</b>                     | Soluble in water, Persistence is unlikely, based on information available.                                                                                  |
| <b>Bioaccumulative Potential</b>       | Bioaccumulation is unlikely                                                                                                                                 |
| <b>Mobility</b>                        | The product is water soluble, and may spread in water systems. Will likely be mobile in the environment due to its water solubility. Highly mobile in soils |
| <b>Endocrine Disruptor Information</b> | This product does not contain any known or suspected endocrine disruptors                                                                                   |
| <b>Persistent Organic Pollutant</b>    | See table for values                                                                                                                                        |
| <b>Ozone Depletion Potential</b>       | This product does not contain any known or suspected substance                                                                                              |

| Component                       | Persistent Organic Pollutant                                                                                                                               | Ozone Depletion Potential |
|---------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------|
| Octanoic acid, pentadecafluoro- | Annex I - Substance subject to prohibitions<br>Annex IV : 50 mg/kg (Waste Management - Conc. Limit)<br>Stockholm Convention - Persistent Organic Pollutant |                           |

## Section 13 - Disposal Considerations

|                                            |                                                                                                                                                                                                                                                                                                                      |
|--------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Waste from Residues/Unused Products</b> | Do not allow into drains or watercourses or dispose of where ground or surface waters may be affected. Wastes, including emptied containers, are controlled wastes and should be disposed of in accordance with all federal, E.P.A., state and local regulations. Assure conformity with all applicable regulations. |
| <b>Contaminated Packaging</b>              | Dispose of this container to hazardous or special waste collection point.                                                                                                                                                                                                                                            |
| <b>Other Information</b>                   | Chemical wastes should be disposed through a licensed commercial waste collection service. Waste codes should be assigned by the user based on the application for which the product was used. Do not empty into drains. Do not flush to sewer. Large amounts will affect pH and harm aquatic organisms.             |

## Section 14 - Transport Information

### IMDG/IMO

|                                |                                          |
|--------------------------------|------------------------------------------|
| <b>UN-No</b>                   | UN3261                                   |
| <b>Proper Shipping Name</b>    | Corrosive solid, acidic, organic, n.o.s. |
| <b>Technical Shipping Name</b> | Octanoic acid, pentadecafluoro-          |
| <b>Hazard Class</b>            | 8                                        |
| <b>Packing Group</b>           | III                                      |

### ADG

|                                |                                          |
|--------------------------------|------------------------------------------|
| <b>UN-No</b>                   | UN3261                                   |
| <b>Proper Shipping Name</b>    | Corrosive solid, acidic, organic, n.o.s. |
| <b>Technical Shipping Name</b> | Octanoic acid, pentadecafluoro-          |
| <b>Hazard Class</b>            | 8                                        |
| <b>Packing Group</b>           | III                                      |

### IATA

|                                |                                          |
|--------------------------------|------------------------------------------|
| <b>UN-No</b>                   | UN3261                                   |
| <b>Proper Shipping Name</b>    | Corrosive solid, acidic, organic, n.o.s. |
| <b>Technical Shipping Name</b> | Octanoic acid, pentadecafluoro-          |
| <b>Hazard Class</b>            | 8                                        |

|                               |                                 |
|-------------------------------|---------------------------------|
| <b>Packing Group</b>          | III                             |
| <b>Environmental hazards</b>  | No hazards identified           |
| <b>Special Precautions</b>    | No special precautions required |
| <b>Additional information</b> | None known                      |

## Section 15 - Regulatory Information

### Safety, health and environmental regulations/legislation specific for the substance or mixture

#### National Regulations      **Australia**

See section 8 for national exposure control parameters.

#### **Standard for the Uniform Scheduling of Medicines and Poisons**

No poison schedule number allocated.

#### **Australian Industrial Chemicals Introduction Scheme (AICIS)**

| Component                                  | Australian Industrial Chemicals Introduction Scheme (AICIS) | Additional information |
|--------------------------------------------|-------------------------------------------------------------|------------------------|
| Octanoic acid, pentadecafluoro- - 335-67-1 | Present                                                     | -                      |

#### **Australian - Illicit Drug Precursors/Reagents Substance List**

This product does not contain any substance(s) on the Illicit Drug Precursors/Reagents list.

#### **Chemicals of Security Concern**

This product does not contain any substance(s) listed on the voluntary National Code of Practice for Chemicals of Security Concern

#### **National pollutant inventory**      Not applicable

#### **Prohibition or notification/licensing requirements**

Shown below are details of specific prohibition/notifications or licensing requirements when they apply.

This product does not contain any substance(s) subject to Prohibition, Authorization or Restriction.

#### **International Inventories**

| Component                       | AICS | NZIoC | EINECS    | ELINCS | TSCA | DSL | NDSL | PICCS | ENCS | ISHL | IECSC | KECL     |
|---------------------------------|------|-------|-----------|--------|------|-----|------|-------|------|------|-------|----------|
| Octanoic acid, pentadecafluoro- | X    | X     | 206-397-9 | -      | X    | -   | -    | X     | X    | X    | X     | KE-27883 |

**Legend:** X - Listed, '-' - Not Listed. SP - Indicates a substance that is identified in a proposed SNUR. **KECL** - NIER number or KE number (<http://ncis.nier.go.kr/en/main.do>)

#### **International Regulations**

#### **Ozone Depletion Potential**      This product does not contain any known or suspected substance



**Persistent Organic Pollutant** See table for values

| Component                                  | Persistent Organic Pollutant                                                                                                                               |
|--------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Octanoic acid, pentadecafluoro- - 335-67-1 | Annex I - Substance subject to prohibitions<br>Annex IV : 50 mg/kg (Waste Management - Conc. Limit)<br>Stockholm Convention - Persistent Organic Pollutant |

**Rotterdam Convention (PIC)** Not applicable

**Basel convention on the control of transboundary movements of hazardous wastes and their disposal**  
Not applicable.

| Component                       | CAS No   | OECD HPV       | Restriction of Hazardous Substances (RoHS) | Seveso III Directive (2012/18/EC) - Qualifying Quantities for Major Accident Notification | Seveso III Directive (2012/18/EC) - Qualifying Quantities for Safety Report Requirements |
|---------------------------------|----------|----------------|--------------------------------------------|-------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------|
| Octanoic acid, pentadecafluoro- | 335-67-1 | Not applicable | Not applicable                             | Not applicable                                                                            | Not applicable                                                                           |

#### Authorisation/Restrictions according to EU REACH

| Component                       | REACH (1907/2006) - Annex XIV - Substances Subject to Authorization | REACH (1907/2006) - Annex XVII - Restrictions on Certain Dangerous Substances                                                            | REACH Regulation (EC 1907/2006) article 59 - Candidate List of Substances of Very High Concern (SVHC)   |
|---------------------------------|---------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------|
| Octanoic acid, pentadecafluoro- | -                                                                   | Use restricted. See item 75.<br>(see link for restriction details)<br>Use restricted. See item 30.<br>(see link for restriction details) | SVHC Candidate list - Toxic for reproduction (Article 57 c)<br>SVHC Candidate list - PBT (Article 57 d) |

After the sunset date the use of this substance requires either an authorization or can only be used for exempted uses, e.g. use in scientific research and development which includes routine analytics or use as intermediate.

<https://echa.europa.eu/authorisation-list>  
<https://echa.europa.eu/candidate-list-table>

## Section 16 - Other Information

### Legend

|                                                                                                      |                                                                                                                              |
|------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------|
| <b>AICS</b> - Australian Inventory of Chemical Substances                                            | <b>NZIoC</b> - New Zealand Inventory of Chemicals                                                                            |
| <b>TSCA</b> - United States Toxic Substances Control Act Section 8(b) Inventory                      | <b>EINECS/ELINCS</b> - European Inventory of Existing Commercial Chemical Substances/EU List of Notified Chemical Substances |
| <b>DSL/NDL</b> - Canadian Domestic Substances List/Non-Domestic Substances List                      | <b>ENCS</b> - Japanese Existing and New Chemical Substances                                                                  |
| <b>IECSC</b> - Chinese Inventory of Existing Chemical Substances                                     | <b>KECL</b> - Korean Existing and Evaluated Chemical Substances                                                              |
| <b>PICCS</b> - Philippines Inventory of Chemicals and Chemical Substances                            | <b>CAS</b> - Chemical Abstracts Service                                                                                      |
| <b>TWA</b> - Time Weighted Average                                                                   | <b>ACGIH</b> - American Conference of Governmental Industrial Hygienists Predicted No Effect Concentration (PNEC)            |
| <b>IARC</b> - International Agency for Research on Cancer                                            | <b>IMO/IMDG</b> - International Maritime Organization/International Maritime Dangerous Goods Code                            |
| <b>ICAO/IATA</b> - International Civil Aviation Organization/International Air Transport Association | <b>ADG</b> Australian Code for the Transport of Dangerous Goods by Road and Rail                                             |
| <b>MARPOL</b> - International Convention for the Prevention of Pollution from Ships                  | <b>OECD</b> - Organisation for Economic Co-operation and Development                                                         |
| <b>NZS 5433:2012</b> - Transport of Dangerous Goods on Land                                          | <b>LC50</b> - Lethal Concentration 50%                                                                                       |
| <b>LD50</b> - Lethal Dose 50%                                                                        | <b>ATE</b> - Acute Toxicity Estimate                                                                                         |
| <b>EC50</b> - Effective Concentration 50%                                                            | <b>RPE</b> - Respiratory Protective Equipment                                                                                |
| <b>WEL</b> - Workplace Exposure Limit                                                                |                                                                                                                              |

**DNEL** - Derived No Effect Level

**POW** - Partition coefficient Octanol:Water

**vPvB** - very Persistent, very Bioaccumulative

**VOC** - (Volatile Organic Compound)

**NOEC** - No Observed Effect Concentration

**BCF** - Bioconcentration factor

**PBT** - Persistent, Bioaccumulative, Toxic

#### Key literature references and sources for data

<https://echa.europa.eu/information-on-chemicals>

Suppliers safety data sheet, Chemadvisor - LOLI, Merck index, RTECS

#### Training Advice

Chemical hazard awareness training, incorporating labelling, Safety Data Sheets (SDS), Personal Protective Equipment (PPE) and hygiene.

Use of personal protective equipment, covering appropriate selection, compatibility, breakthrough thresholds, care, maintenance, fit and standards.

First aid for chemical exposure, including the use of eye wash and safety showers.

**Revision Date** 17-Nov-2022

**Revision Summary** Not applicable.

**This Safety Data Sheet (SDS) is prepared in accordance to and complies with the requirements of Safe Work Australia - Work Health and Safety Regulations (WHS Regulations).**

#### Disclaimer

The information provided in this Safety Data Sheet is correct to the best of our knowledge, information and belief at the date of its publication. The information given is designed only as a guidance for safe handling, use, processing, storage, transportation, disposal and release and is not to be considered a warranty or quality specification. The information relates only to the specific material designated and may not be valid for such material used in combination with any other materials or in any process, unless specified in the text

## End of Safety Data Sheet